25900670 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course(VAC) - MG3VACTXLA02 - WET PROCESSING TECHNIQUES FOR TEXTILES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. State the condition of fabric during drying in the tentering process. [K] / [CO1]

   a). Folded and compressed b). Rolled onto a spool

   c). Held in the stretched position d). Immersed in dye

2. State the reason why sizing or waxy substances should be removed before singeing. [K] / [CO1]

   a). To make the fabric shiny b). To prevent scorching and burning

   c). To add strength to the fabric d). To keep the rollers clean

3. State the key difference between waterproof and water-repellent fabrics. [K] / [CO2]

   a). Waterproof fabrics absorb water, water-repellent fabrics do not b). Water-repellent fabrics block all water, waterproof fabrics do not

   c). Waterproof fabrics block all water, while water-repellent fabrics resist light rain d). There is no difference

4. State a key drawback of fibre dyeing. [K] / [CO3]

a). Poor penetration b). Requires early color decision

c). Limited use in wool fabrics d). Weak fastness properties

5. State the reason why curing is applied after hydro-extraction in tumble dyeing. [K] / [CO3]

a). To dry the garment completely b). To fix the dye by heat application (e.g., ironing or steam pressing)

c). To rinse out excess dye d). To shrink the garment intentionally

6. State the method between winch and jet dyeing that consumes less water, energy, and chemicals. [K] / [CO3]

a). Winch dyeing b). Jet dyeing

c). Both equally d). None

7. State the fibres dyed with disperse dyes. [K] / [CO3]

a). Cotton b). Polyester

c). Wool d). Jute

8. State the process involved in naphthol dyes. [K] / [CO3]

a). Oxidation process b). Coupling reaction

c). Reduction process d). Mordanting process

9. Identify the fabric most suitable for discharge printing. [K] / [CO4]

a). Polyester b). Wool

c). Cotton dyed with reactive dyes d). Acrylic

10. Why is a separate engraved roller required for each color in roller printing? [K] / [CO4]

a). To reduce machine wear b). To ensure accurate alignment of multiple colors

c). To simplify rinsing d). To increase drying speed

[1x10 = 10]

Part B

Fill in the Blanks Questions

Answer all questions

Each question carries **1** marks

11. Sodium chlorite is used for bleaching ..... and protein fibers. [K] / [CO1]

12. Shearing enhances the fabric's ..... by removing lint and uneven surfaces. [U] / [CO1]

13. Shrink resistance treatment is often used to prevent shrinkage during ..... [K] / [CO2]

14. In pad dyeing, fabric in open width passes through a dye bath followed by ..... rollers. [U] / [CO3]

15. Acid dyes are..... soluble. [K] / [CO3]

16. Basic dyes are ..... charged. [K] / [CO3]

17. One major disadvantage of DP finish is the release of....., which can raise health concerns. [U] / [CO2]

18. ..... gives a fabric a uniform solid shade even when made of different fibers. [U] / [CO3]

19. One limitation is that rotary screen printing is not economical for ..... orders due to its high setup costs. [U] / [CO4]

20. Duplex printing prints on ..... sides of the fabric. [K] / [CO4]

[1x10 = 10]

Part C

Answer in One Word or One Sentence

Answer all questions

Each question carries **1** marks

21. Name the one-word process that enhances both strength and luster of cotton. [K] / [CO1]

22. Name one natural sizing agent. [K] / [CO1]

23. Name one industry where flame retardant fabrics are crucial. [K] / [CO2]

24. Define direct dyes. [K] / [CO3]

25. Mention one property of reactive dyes. [K] / [CO3]

26. State the alternative name for skein dyeing. [K] / [CO3]

27. Identify the dyeing method that ensures the most uniform dye distribution in wool fibres. [K] / [CO3]

28. State the reason gentle motion (swinging or stirring) is used during the dip-dye process. [K] / [CO3]

29. State one factor that makes block printing sustainable. [K] / [CO4]

30. Jig dyeing is mainly used for which type of textiles? [U] / [CO3]

[1x10 = 10]

Part D

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

31. State the physical changes cotton fibers undergo during mercerization. [K] / [CO1]

32. Define desizing and state its importance in textile processing. [K] / [CO1]

33. Explain how chlorination helps in wool shrink-resistance. [U] / [CO2]

34. Define garment dyeing. [K] / [CO3]

35. State any two advantages of vat dyes. [K] / [CO3]

36. Differentiate textile printing from dyeing. [U] / [CO4]

[2x5 = 10]

Part E

Long Answer Questions

Answer any **2** questions

Each question carries **5** marks

37. Discuss at least three purposes of flocking and the benefits each provides to fabric. [U] / [CO1]

38. Differentiate between natural dyes and synthetic dyes. [U] / [CO3]

39. Define fabric dyeing and explain its importance as a stage in textile production. [U] / [CO3]

40. Explain the main components and step-by-step process of screen printing. [U] / [CO4]

[5x2 = 10]